



KOLEJ MATRIKULASI MELAKA

COMPUTER SCIENCE 1
IQUA
DC014
SEMESTER I, SESSION 2024/2025

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NAME : _____
MATRIC NO. : _____
TUTORIAL : _____

Answer all questions.

1 Qhairullah works as an advertising designer for FuzzyFalza Company, a global beverage company launching a new line of energy drinks targeted at young adults aged 18-30. He needs to create a captivating advertising campaign that uses multimedia elements to capture the attention of the audience and drive brand awareness and sales.

(a) Suggest **one (1)** suitable multimedia element that he can use to advertise the campaign using social media platforms.

[1 mark]

(b) Based on your answer in **1(a)**, justify the purposes of the chosen multimedia element.

[2 marks]

(c) Qhairullah was also asked by his company to make an advertisement in the latest representation of multimedia applications in four dimensions.

(i) Explain the dimension of four dimensional.

[2 marks]

(ii) State other types of multimedia applications representation that can be used by him to achieve the company's goals.

[1 mark]



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2 Multimedia design involves the creation of visually and interactively engaging content that combines various types of media elements to convey information, entertain, or communicate a message.

(a) Identify the phases of multimedia design and activities that involved based on the scenario given.

[6 marks]

(i) Asyiela Putri contributed her voice for the characters of Upin and Ipin.

Phase : _____
Activity : _____

(ii) BillyMilly's production company uses multiple platforms to ensure that the multimedia projects it develops can be played on all platforms.

Phase : _____
Activity : _____

(iii) Najah led the multimedia development team in identifying the content, cost and duration of the project to be developed.

Phase : _____
Activity : _____

(b) (i) Based on **FIGURE 1(a)** and **FIGURE 1(b)**, state the authoring tools below.

[2 marks]



FIGURE 1(a)

Types of authoring tool: _____



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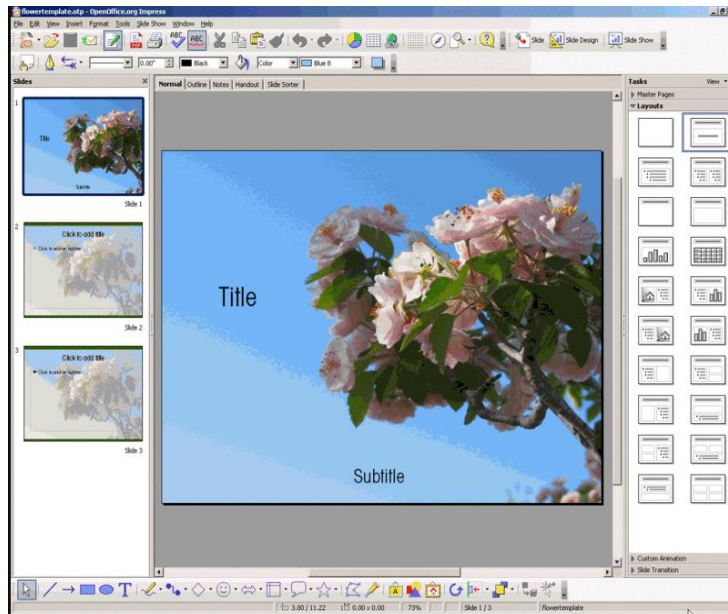


FIGURE 1(b)

Types of authoring tool: _____

- (ii) Differentiate **FIGURE 1(a)** and **FIGURE 1(b)** based on the technique criteria.

[2 marks]

FIGURE 1(a)	FIGURE 1(b)



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- (c) Based on the **FIGURE 2**, explain the type of navigation **Z** used in Google Maps.

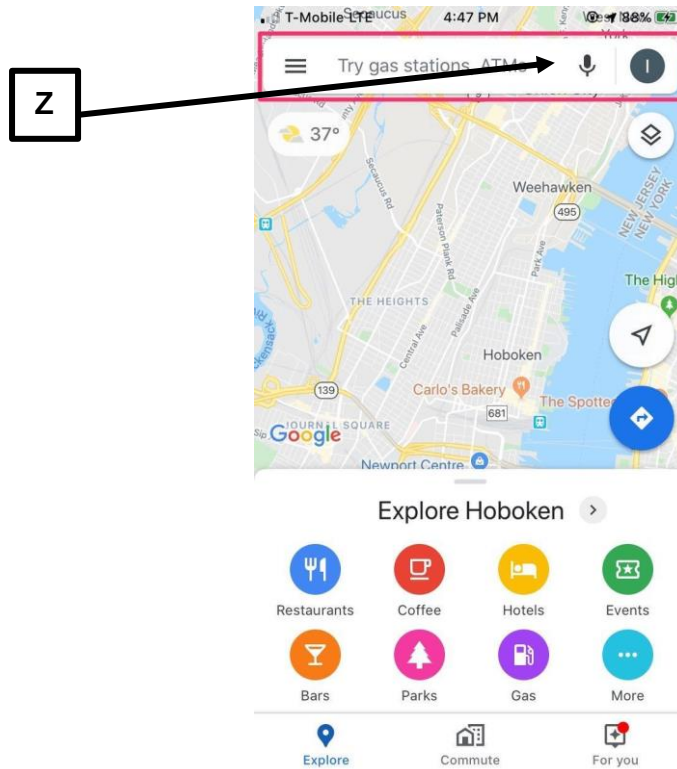


FIGURE 2

[2 marks]

- (d) A hard disk, often referred to simply as a "hard drive" or "hard disk drive" (HDD), is a non-volatile storage device used to store and retrieve digital data. Describe **one (1)** characteristic of using offline as a medium of delivery based on its capacity and speed of transfer rate.

[4 marks]

	Characteristic
Capacity	
Speed data of transfer rate	



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- 3 In today's digital era, databases play a critical role in the success and efficiency of businesses across various industries.

- (a) Differentiate between file processing systems and the database system.

[4 marks]

File Processing System	Database System

- (b) Based on the scenario below, identify **two (2)** disadvantages of database system in Rayqal's scenario.

'Rayqal is the IT manager of a growing e-commerce company. His team is considering transitioning from storing data in spreadsheets to implementing a full-fledged database system to manage customer orders, inventory, and sales data.'

[2 marks]



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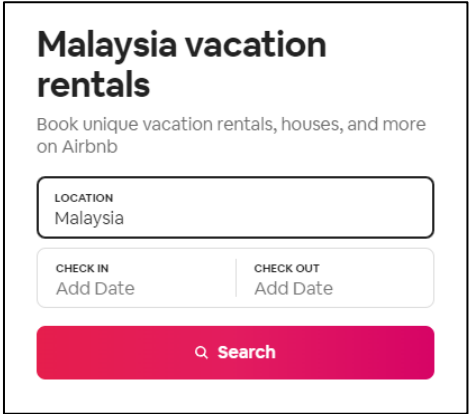
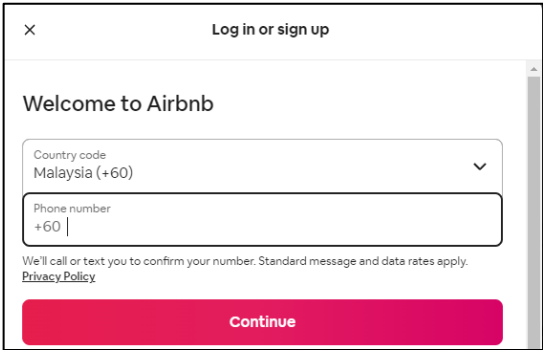
- 4 Modern database systems are optimized to expand both horizontally and vertically, meeting the rising data volumes and increasing user demands.

(a) Explain data integrity in a database system.

[2 marks]

(b) Many websites do utilize some form of database to manage and organize their content. Airbnb's homepage allows users to search their destination, check-in and check-out date. Identify the correct database object based on the image given below.

[2 marks]

	Image	Database Object
(i)		
(ii)		



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- 5 Networking is important in virtually every industry today due to the increasing reliance on digital technologies for communication, data sharing, and collaboration.

(a) Describe the following uses of network.

(i) Global Positioning System (GPS).

[2 marks]

(ii) Television and Radio Broadcasting.

[2 marks]

- (b) Networking media play a crucial role in facilitating communication between devices within a network, enabling data transmission and connectivity across various physical and virtual environments.

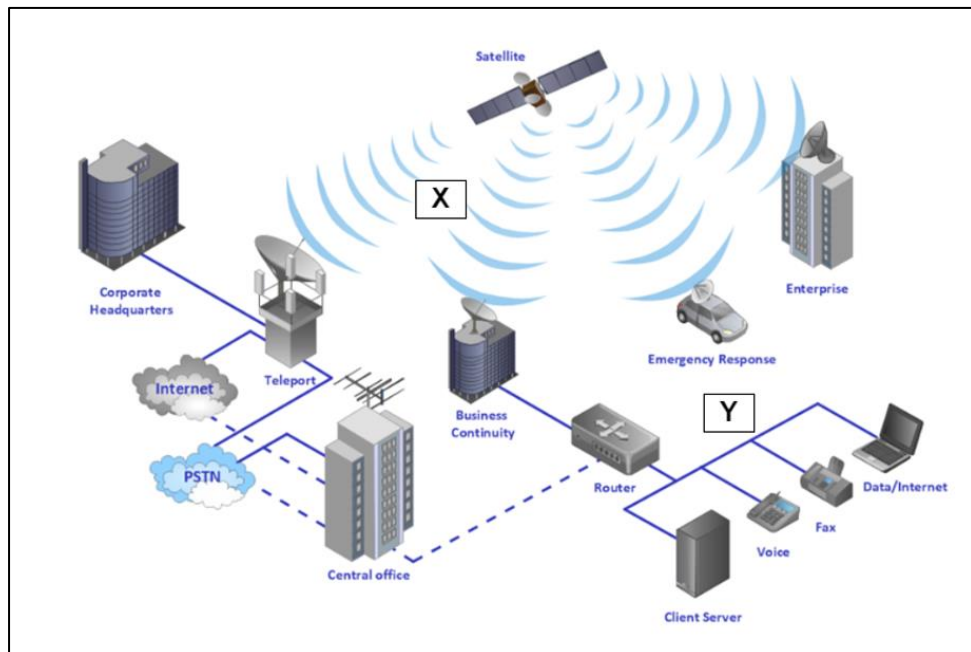


FIGURE 3

- (i) Identify the type of networking media based on the **FIGURE 3**.

[2 marks]

X : _____
Y : _____



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- (ii) Explain **one (1)** difference between networking media X and Y.

[4 marks]

Networking Media X	Networking Media Y

- (iii) Give **two (2)** examples of networking media X.

[2 marks]

- (c) Based on **FIGURE 4**,

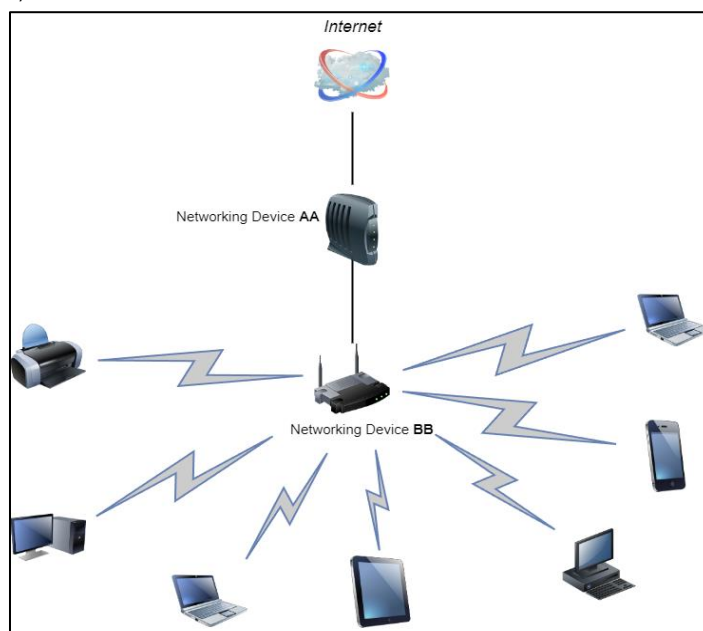


FIGURE 4

- (i) Identify the type of networking device.

[2 marks]

AA : _____
BB : _____



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- (ii) Based on the **FIGURE 4**, identify the type of networking media and give an example that is suitable for connecting the devices to the Networking Device **BB**.

[2 marks]

Type:		Example:	
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- (d) Identify the example of networking device based on the scenario below.

[2 marks]

	Scenario	Example of Networking Device
(i)	Zaharith is in his large apartment, and the Wi-Fi signal in his bedroom is weak. To fix this, he installs a device with antennas in the hallway that helps boost the signal, allowing him to stream videos without interruptions.	
(ii)	Danish is managing the Wi-Fi network at a university. To ensure students have wireless internet in all classrooms, he installs devices that create additional wireless coverage zones by connecting to the main network.	



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6 The process of using the Internet to find and acquire information from the many resources accessible on the World Wide Web is referred to as "web searching," or "internet searching.

(a) The owner of a bookstore wants to know which book has been the best-seller this year. You use the computer to search online bookstores, review websites, and best-seller lists to identify the most popular book among readers.

(i) Determine which kind of search tool is best to utilize in this situation.

[1 mark]

(ii) Describe search tool is that stated in 6(a)(i).

[2 marks]

(iii) Give **one (1)** example of the related search tool.

[1 mark]

(b) Explain **one (1)** advantage of the search tools mentioned in question 6(a).

[2 marks]

(c) Internet search strategies are techniques used to efficiently locate relevant information on the World Wide Web. Identify the most suitable search strategies based on given scenario.

[3 marks]

	Scenario	Search Strategies
(i)	Grace work in a healthcare setting and need to research a specific tumor called "Cloud Cytoma" to provide information to patients.	
(ii)	Alex is looking for job opportunities in the marketing field. He is interested in positions that focus on either social media marketing or email marketing. He might search for "marketing job social media OR email" to find job listings that match either criteria.	
(iii)	Sarah is conducting research for her psychology paper and wants to find articles. Sarah might enter "author:John Smith psychology" to focus her search on articles written by John Smith within the field of psychology.	



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- (d) Web activities refer to the wide range of actions and interactions that individuals engage in while using the World Wide Web.

Name the types of web activities based on the statements given below.

[5 marks]

	Statement	Types of Web Activities
(i)	Discussion platforms where users can post messages, ask questions, share information, and engage in conversations with other users on specific topics or interests.	
(ii)	Streaming services like Netflix, Hulu, Amazon Prime Video, and Disney+ have transformed the way people consume movies, TV shows, and documentaries. These platforms offer on-demand access to a vast library of content, allowing users to watch their favorite movies and shows anytime, anywhere, on any device.	
(iii)	This web activities have transformed the way people shop and conduct business, offering convenience, accessibility, and a global marketplace at their fingertips.	
(iv)	These platforms often include learning management systems (LMS), video conferencing tools, interactive multimedia content, and online assessments, providing a rich and immersive learning experience.	
(v)	These platforms provide powerful tools for networking, self-expression, and community building, allowing users to interact with friends, family, and peers, as well as with brands, organizations, and influencers, across the globe.	

- (e) Web privacy refers to the right of individuals to control the collection, use, and sharing of their personal data and information when accessing websites, applications, and online services.

- (i) State **two (2)** ways to improve web browsing privacy.

[2 marks]



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- (ii) Describe e-mail privacy.

[2 marks]

- (f) The Internet of Things, or IoT, is a big change in technology. It means regular stuff connects to the internet, so they can share data on their own. This includes things like gadgets, cars, and appliances with sensors and internet connections.

Suggest **one (1)** impact of IoT in health care.

[2 marks]

END OF QUESTIONS

Endorsed by:

NOR ILYANI BINTI OTHMAN
Ketua Unit Sains Komputer
Kolej Matrikulasi Melaka
78300 Masjid Tanah, Melaka

